# Alex Gold

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Mechanical design engineer shipping high-voltage charging hardware into vehicle production at Tesla. Five years of hands-on work across battery systems, power electronics, motor control and mechatronics, from cell chemistry to production tooling. Focused on electromechanical systems that sense and respond to the physical world.

## Experience

**Tesla** — Mechanical Design Engineer, High Voltage Distribution · Palo Alto, CA · June 2024 – Present

- Design owner of multiple parts, including an actuator system developed from an existing design into a more reliable product; take parts from early prototypes through ramp to manufacturing at massive scale.
- Designed moving plastic and metal assemblies in CATIA V6 (3DX) to automotive high-reliability standards; owned tolerance stack-ups, GD&T and released 2D drawings.
- Drove DFM with overseas suppliers through tooling iterations; supported initial production builds on the factory floor.
- Built validation test suites and used failure-mode analysis and test data to bound reliability risk before launch.
- Defined early high-voltage distribution layout for a new vehicle program: cable routing, chargeport integration and parametric models used across teams for mass reduction.

**Lion Energy** — Mechanical Engineering Intern · American Fork, UT · May – Sep 2023

- Designed and built the first prototypes of a 4.5 kWh, 50 V LFP modular home-energy storage pack; delivered 3 working units.
- Ran thermal FEA to set the cooling strategy and structural FEA to size the enclosure, handles and rack.
- Released drawings and sourced parts from 12+ vendors; presented in design reviews with electrical, software and validation teams.

**BYU Formula SAE Electric** — Accumulator Lead (Capstone), Founding Member · Provo, UT · Aug 2022 – Apr 2024

- Led a 10-engineer team designing a 450 V, 8 kWh accumulator from Tesla-sponsored 21700 cells; owned cell layout, fused collector plates, BMS integration and thermistor isolation. Pack passed FSAE technical inspection.
- Wrote MATLAB fuse-sizing models and built a DAQ test rig to validate them; helped laser-weld modules.
- As a founding member, set pack voltage and current, selected motor and inverter through decision matrices, imported components and brought up the first motor spin.

**BYU Lithium-Ion Battery Lab** — Research Assistant · Provo, UT · Sep 2022 – Apr 2024

- Built a continuous process converting halloysite mining waste into silicon anode material; rebuilt a failed 250 °C, 5 bar extruder design as a peristaltic-pump system with software-controlled valves.
- Fabricated single-layer pouch cells from experimental material and ran charge/discharge cycling.
- Modified pressure vessels for in-situ lithium-air cell testing on a US Army-funded program.

## Projects

**Haptic Desk Turntable** — personal, in progress · [demo link]

- Scratchable 2-inch platter driven by a BLDC gimbal motor under field-oriented control (SimpleFOC, AS5600 encoder), pushing torque back against the user's hand.
- ESP32-S3 firmware isolates the 10–20 kHz motor loop on one core; the other runs a PSRAM ring buffer with variable-speed, reversible playback, an I2S DAC and Bluetooth audio and call profiles.
- Validated scratch feel in a browser simulation before building hardware; sized the motor and 5 V USB-C power budget from first principles.

**Thermal Printer Camera** — personal, lead designer, in progress

- Own the full product design of a compact instant-print camera, partnering with a firmware engineer; designing toward a small production run.

**Autonomous Bike Chain Lubricator** — BYU mechatronics capstone, solo

- Clip-on device that cleans and lubricates a chain: stepper drive, peristaltic pump, microcontroller, boost converters, Li-ion charging and an animated display UI.

## Education

**Brigham Young University** — B.S. Mechanical Engineering · Provo, UT · April 2024

Coursework: Mechatronics, Power Electronics, Heat Transfer, Materials Engineering

## Skills

- **Mechanical:** CATIA V6 (3DX) and V5, SolidWorks; GD&T, tolerance analysis, DFM for injection-molded and metal parts
- **Analysis:** ANSYS and SolidWorks FEA (structural, thermal), MATLAB, LTspice
- **Electronics & firmware:** C/C++, ESP32 and Arduino, FOC motor control, I2S/I2C, DAQ, power electronics
- **Fabrication:** rapid prototyping, 3D printing, CNC and manual machining, laser welding, soldering